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Interstellar Shocks: Principles

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ABSTRACT

We will look into supersonic dynamics following the derivation given by Kwok (2007). A brief discussion of magnetic field is from Dopita & Sutherland (2003).

1. INTRODUCTION

When high-spped gas runs into thee general ISM, they cause a distrubance to the ISM. The resultant changes in velocity, density, pressure, and temperature are communicated downstream at the soud speed. But for a supersonic gas, the motion itself is faster than the speed of communication, and instead of a smooth transition, these physical quantities undergo a sudden change in values over a small distance. This phenomenon is referred to as a *shock*.

2. THE RANKINE-HUGONIOT JUMP CONDITIONS

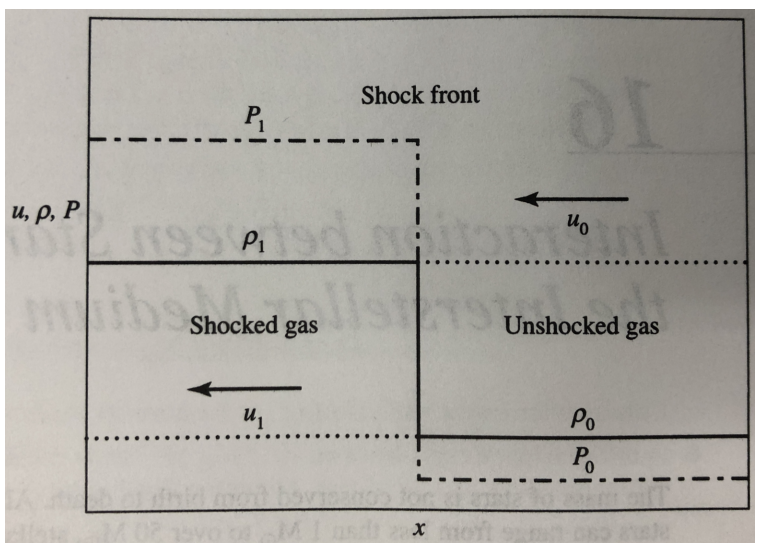

Figure 1. A schematic diagram illustrating the sudden change in velocity, density, and pressure after the passage of a shock (in shock-fixed coordinates). Fig 16.1 of Kwok (2007).

Suppose that the pre-shock gas is at rest, the post-shock gas has a velocity of v_1 and the shock velocity is v_s. If we define the speed of the gas in the reference frame of the shock as u, then the gas speed in front (u_0) and behind (u_1) are $u_0 = -v_s$ and $u_1 = v_1 - v_s$. Since the mass flux entering the shock from in front must be the same as the mass flux leaving the shock, we have

$$\rho_0 u_0 = \rho_1 u_1 \tag{1}$$

Similarly, the change in momentum flux ($(\rho u)u$) must be balanced by the pressure change across the front

$$P_0 + \rho_0 u_0^2 = P_1 + \rho_1 u_1^2 \tag{2}$$

For an ideal gas, the energy density is the sum of the kinetic energy $1/2\rho u^2$ and internal energy $U = \mathcal{U}\rho$, where the energy per unit mass is

$$\mathcal{U} = \frac{P}{(\gamma - 1)\rho} \tag{3}$$

The difference between the rate of energy enetering the leaving gthe shock is the work done by the gas as a result of the pressure differnce across the shock:

$$u_1\left(\frac{1}{2}\rho_1 u_1^2 + U_1\right) - u_0\left(\frac{1}{2}\rho_0 u_0^2 + U_0\right) = P_0 u_0 - P_1 u_1 \tag{4}$$

Substituting Eq. 1 and 3 into Eq. 4, we have

$$\frac{1}{2}u_0^2 + \frac{\gamma}{\gamma-1}\frac{P_0}{\rho_0} = \frac{1}{2}u_1^2 + \frac{\gamma}{\gamma-1}\frac{P_1}{\rho_1} \tag{5}$$

Define trhe *Mach number* as $M_0 = |u_0|/c_{s0}$, where $c_{s0}^2 = \gamma P_0/\rho_0$, then we have

$$P_0 = \frac{u_0^2\rho_0}{\gamma M_0^2} \tag{6}$$

Substituting Eq. 1, 2 and 6 into Eq. 5, we have

$$\begin{aligned} ax^2 + bx - c &= 0 \\ x &= \frac{\rho_0}{\rho_1} \\ a &= \frac{1}{2} - \frac{\gamma}{\gamma-1} \\ b &= \frac{\gamma}{\gamma-1} + \frac{1}{(\gamma-1)M_0^2} \\ c &= \frac{1}{2} + \frac{1}{(\gamma-1)M_0^2} \end{aligned}$$

The two roots of this quadratic equation represent the densities on either side of the shock:

$$\frac{\rho_0}{\rho_1} = \frac{-\left[\frac{\gamma}{\gamma-1} + \frac{1}{(\gamma-1)M_0^2}\right] \pm \frac{M_0^2-1}{M_0^2(\gamma-1)}}{1 - \frac{2\gamma}{\gamma-1}}$$

The root with the negative sign gives back the pre-shock density ($\rho_0 = \rho_1$), whereas the root with the positive sign gives

$$\frac{\rho_1}{\rho_0} = \frac{(\gamma+1)M_0^2}{(\gamma+1) + (\gamma-1)(M_0^2-1)} \tag{7}$$

For a strong shock, $M_0 \to \infty$:

$$\frac{\rho_1}{\rho_0} \approx \frac{\gamma+1}{\gamma-1} \tag{8}$$

If the shock is adiabatic ($\gamma = 5/3$), the degree of compression cannot exceed 4, no matter how strong the shock is. An adiabatic gas, not being able to radiate away the excess energy, will have all the kinetic energy changed into internal energy, which resists further compression.

For an isothermal shock ($\gamma = 1$) where all excess energy is effectively radiated away, no such finite limit exists and $\rho_1/\rho_0 \to \infty$ as $M_0 \to \infty$.

From Eq. 1 and 7, we have the change of gas velocity across the shock:

$$\frac{u_0}{u_1} = \frac{(\gamma+1)M_0^2}{(\gamma+1) + (\gamma-1)(M_0^2-1)} \tag{9}$$

Transforming this back to the stellar frame of reference using $u_0 = -v_s$ and $u_1 = v_1 - v_s$, we have

$$\begin{aligned} \frac{-v_s}{v_1 - v_s} &= \frac{(\gamma+1)M_0^2}{(\gamma+1) + (\gamma-1)(M_0^2-1)} \\ v_1 &= \left(\frac{2}{\gamma+1}\right) v_s \end{aligned} \tag{10}$$

For a strong adiabatic shock, the shocked gas is accrelerated to $3/4v_s$.

For an isothermal shock, Eq. 9 gives $u_0 = M_0^2 u_1$. In the stellar frame of reference

$$\begin{aligned} -v_s &= M_0^2(v_1 - v_s) \\ v_1 &= \left(1 - \frac{1}{M_0^2}\right) v_s \end{aligned} \tag{11}$$

For a strong shock, the shocked gas is accrelerted to the speed of the shock.

We can substitute the results from Eq. 6, 7 and 9 back to Eq. 2 to obtain the jump in pressure:

$$\frac{P_1}{P_0} = \frac{2\gamma}{\gamma+1} M_0^2 - \frac{\gamma-1}{\gamma+1} \tag{12}$$

Eq. 7, 9 and 12 are together referred to as the jump condidtions across a shock.

3. SHOCK HEATING

Using the ideal gas law $P = \rho kT/(\mu m_p)$, we can derive the change in temepratures:

$$\frac{T_1}{T_0} = \frac{[(\gamma+1) + 2\gamma(M_0^2 - 1)][(\gamma+1) + (\gamma-1)(M_0^2-1)]}{(\gamma+1)^2 M_0^2} \tag{13}$$

For a strong shock:

$$T_1 = \frac{2\gamma(\gamma-1)}{(\gamma+1)^2} M_0^2 T_0 = \frac{2\gamma(\gamma-1)}{(\gamma+1)^2} \frac{u_0^2 \rho_0}{\gamma P_0} \frac{\mu m_p P_0}{\rho_0 k} = \frac{2(\gamma-1)}{(\gamma+1)^2} \frac{\mu m_p}{k} u_0^2 \tag{14}$$

When $\gamma = 5/3$,

$$\begin{aligned} T_1 = \frac{3}{16} \frac{\mu m_p}{k} v_s^2 &= 2890\,\mathrm{K} \left(\frac{\mu}{1.273}\right) \left(\frac{v_s}{10\,\mathrm{km\,s^{-1}}}\right)^2 \\ &= 1.38 \times 10^7\,\mathrm{K} \left(\frac{\mu}{0.609}\right) \left(\frac{v_s}{1000\,\mathrm{km\,s^{-1}}}\right)^2 \end{aligned}$$

For a shock velocity of $1000\,\mathrm{km\,s^{-1}}$, the post-shock gas can be heated up to over 10^7 K. Shocks therefore represent an effective heating mechanism of the ISM.

4. SHOCKS WITH MAGNETIC FIELDS*

When there is a (transverse) magnetic field present in the preshock medium, the pressure due to the transverse component of the magnetic field is likely to dominate the gas pressure at some point in the postshock plasma, since the magnetic pressure increases as the square of the density, but the gas pressure is limited by the ram pressure of the material entering the shock. The implication of this is that, at a certain point in the cooling plasma, the shock switches from being gas-pressure supported to being supported by the magnetic pressure. If the magnetic field pressure in the postshock gas dominates the gas pressure and the ram-pressure terms, and if the gas and magnetic pressures in the preshock gas are negligible compared with the ram-pressure (the fast shock limit) then, the momentum equation becomes:

$$\frac{B_1^2}{8\pi} = \rho_0 v_s^2 \tag{15}$$

Furthermore, if the magnetic field is also frozen in the flow, then equating the preshock ram pressure to the postshock transverse magnetic pressure, we obtain the maximum compression factor attainable in the shock:

$$\frac{\rho_1}{\rho_0} = \sqrt{2} M_A \tag{16}$$

where the *Alfven Mach number* M_A is defined as the ratio of the shock velocity to the Alfven velocity defined by

$$\frac{B^2}{8\pi} = \frac{1}{2} \rho v_A^2 \tag{17}$$

REFERENCES

Dopita, M. A., & Sutherland, R. S. 2003, Astrophysics of the diffuse universe

Kwok, S. 2007, Physics and Chemistry of the Interstellar Medium